

Reverse physics and pure-Weyl gravity

Executive summary for physicists

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Research context. Asger Alstrup Palm, a computer scientist and Honorary Professor at DTU Compute, orchestrates the programme and is the accountable human contact. AI systems perform substantial derivation, programming, literature work, adversarial review, and drafting. The work should be judged from its arguments, claim boundaries, and reproducible evidence—not accepted or rejected because of its unusual authorship.

Ninety-second summary

This repository now contains two connected research programmes.

The first asks whether pure Weyl gravity, and related fourth-order theories, can survive a sequence of classical, nonlinear, causal, quantum, and empirical tests. It has produced exact constructions and scoped obstructions, but no positive complete quantum theory, scattering theory, or phenomenologically validated replacement for general relativity.

The second asks a more general question: what part of a physical theory comes from physical postulates, what part comes from the chosen mathematical carrier, and what part comes from logic, infinity, or existence principles? Its central judgement is

$$L + S + M + \text{Enc}(P) \vdash O,$$

where logic L , set or existence theory S , mathematical carrier M , and encoded physical postulates P are kept separate from the obligation O being proved. The resulting **reverse-physics atlas** has six foundational regimes, six carrier families, and sixteen physical obligations: 576 navigational coordinates. These are not 576 theories, and cell coverage does not imply that the ingredients compose.

Several recent results sharpen the combined picture:

- the authoritative classical pure-Weyl BV snapshot has passed its independent Gate-A import decision;
- typed nonlinear q_2/q_3 Green compatibility and a full 386-row BRST–Hadamard two-point **pseudo-state pair** are certified;
- the pseudo-state is indefinite, not a positive quasifree state or physical graviton Hilbert space;
- a bounded NGC 3198 comparison gives Mannheim conformal gravity the smaller unweighted residual, but GR plus an NFW halo the better weighted fit and the only passing random-error gate among the three galactic models tested;
- an explicit Bateman–Turok Euclidean torus family refutes a tempting deterministic all-field stability bound at the free spectral scale;
- eight end-to-end theory passports and nine programme prototypes now expose where mathematical pieces exist but the joins to predictions or data remain open.

The headline is therefore not that one unconventional theory has won. It is that assumptions, carriers, proofs, interfaces, and empirical tests can be made separately reviewable.

What is established—and what is not

Established in declared settings

- Positive, Krein, vacuum, and Jordan completions of selected fourth-order systems, together with exact interaction obstructions.
- Residual cohomology and causal BV–BFV transport for free pure-Weyl gravity on the conformal cylinder in a selected closed-universe polarization.

- Additional compact axial and polar classical directions with nonzero action-derived pairings; their signs do not follow the shortcut “Einstein positive, additional negative.”
- A nonzero strict local Euclidean one-loop BV anomaly. A formal shifting compensator restores the tested local Euclidean QME at one loop, but changes the theory and counterterm space.
- Axial Schwarzschild endpoint non-selection, a non-split repeated Regge–Wheeler extension, one validated defective $\ell = 2$ QNM, and a nonzero second-order pole of the compactly cut-off radial Green operator.
- An immutable classical import snapshot and independent Gate-A verification of its declared nilpotency, contraction, chain-map, cyclicity, residual cohomology, and pairing obligations.
- Full-complex BRST–Hadamard two-point pseudo-state data on the 386-row off-shell carrier, including the declared bisolution, graded CCR, wavefront, Ward, reality, and stationarity conditions.
- Exact finite, constructive, weak-foundation, and Krein examples that show how particular basis-selection or completion assumptions can be avoided in stated scopes.
- Scoped empirical controls: standard GR passes the declared Cassini gate; GR+NFW passes the bounded NGC 3198 random-error gate.

Partial, changed-theory, or interface-local

- Positive metrics and probability rules on selected finite or reduced blocks.
- Relational clocks and detector preparations on compact laboratories.
- Finite-harmonic nonlinear continuation and several exact obstruction branches, without a complete bounded nonlinear cone.
- A compensator restoration in an enlarged theory, not strict pure Weyl.
- A Hadamard pseudo-state pair without positivity on physical cohomology.
- A mode-reduced retarded-to-resonance bridge without a complete global metric/Bondi waveform theorem.
- Matrix cells and programme prototypes whose selected ingredients have not been joined by typed composition theorems.
- Two Lean/Physlib proof passports: useful independent formal replay surfaces, but not stronger physics claims or upgraded matrix grades.

Open

- A positive full-BV physical state, physical Hilbert space, scattering, and unitarity.
- Renormalized Lorentzian time-ordered products, causal pAQFT, and a Lorentzian quantum master equation.
- Quantum residual transfer and a particle interpretation of residual cohomology. The centered classes $[W_+^2]$ and $[W_-^2]$ are certified deformation/vertex classes, not one-particle gravitons.
- A complete Lorentzian off-shell BV propagator represented by effective distribution kernels.
- Population-level, held-out, systematics-aware comparison of the galactic models.
- The Bateman–Turok Witten quotient, Gibbs typicality, interacting H^{-1} moment, continuum measure, and Osterwalder–Schrader reconstruction.
- Full Schwarzschild contour deformation, real-source excitation, bounded metric/Bondi reconstruction, and polar or all-multipole analogues.
- A weakest-foundation theorem for a full physical theory.

Claim architecture

Every quantum result carries one or more dependency tags:

LOCAL–ALGEBRAIC
 EUCLIDEAN–SPECTRAL
 REDUCED–MODE
 LORENTZIAN–CAUSAL

A reduced-mode calculation is not evidence for a Lorentzian-causal theorem. A Euclidean determinant is not a Lorentzian state. A cohomology class is not a particle. A finite exact model is not a continuum limit.

Quantum lifecycle states are also distinct:

CLASSIFIED
COEFFICIENT_COMPUTED
QME_RESTORED
RESIDUAL_TRANSFERRED
LORENTZIAN_CERTIFIED

The programme deliberately permits later stages to remain visible when an earlier interface is open. That is diagnostic information, not permission to compose around the missing step.

Five scientific strands

1. Fourth-order completion and the ghost test

Papers 01–06 distinguish a positive finite-dimensional metric, a Krein real form, vacuum covariance, field representation, Fock sector, the equal-frequency Jordan limit, and interaction stability. These objects are related but not interchangeable. A local negative sign does not by itself prove a negative probability particle; conversely, a positive metric on a free block does not guarantee a compatible interacting theory.

Papers 07–18 carry this separation into pure-Weyl gauge theory, compact model universes, Schwarzschild end-points, and black-hole thermodynamics. The stable Phase-1 conclusion is a classification: additional classical directions survive some reductions, while global balance, resonance, stability, and quantum tests obstruct others. No tested branch passes every level.

The sharpest black-hole case remains valuable but is no longer the programme’s sole front door. In axial Schwarzschild $\ell = 2$, ordinary future-horizon and one-sided radiation conditions do not select the Einstein subsector. A repeated spin-two Regge–Wheeler pair forms a non-split self-extension. At one validated damped frequency it has a length-two root chain and a rank-one second-order cut-off Green pole. The local contour contains a $te^{i\omega_n t}$ term, but no complete causal detector waveform has been proved.

2. Classical authority, quantum anomaly, and Hadamard data

The classical BV–BFV complex is the authoritative input to the quantum programme. Gate A imports a content-addressed snapshot rather than rebuilding a competing complex in the quantum tree. The snapshot passes the declared algebraic, homological, cyclic, residual, and pairing checks.

Post-freeze work establishes typed q_2/q_3 compatibility with advanced and retarded Green homotopies and then constructs a BRST–Hadamard two-point pair on the full 386-row carrier. The word **pseudo-state** is load-bearing. The pair is indefinite and does not establish positivity on physical cohomology, a unique zero-mode policy, a Feynman propagator, renormalized products, causal pAQFT, QME restoration, particle space, scattering, or unitarity.

This Lorentzian frontier is logically separate from Paper 12’s local anomaly classification. Strict pure Weyl is locally QME-obstructed at one loop in the declared Euclidean quotient. A formal compensator extension restores a changed local theory at that order. Neither result is a positive Lorentzian quantum theory.

3. Mannheim conformal gravity and bounded data contact

Papers 18–19 isolate what the static Mannheim–Kazanas explanation assumes. The linear potential is forced by the radial biharmonic equation, but its physical interpretation depends on matter, conformal frame, source moments, and boundary data. In the simplest constant-scalar frame the linear term is removed and massive particles follow the conformally rescaled geometry.

The common NGC 3198 protocol compares 39 points and quoted random errors:

- Newtonian baryons: RMS 23.896 km/s, $\chi^2_\nu = 128.722$ — fail.
- GR+NFW: RMS 5.148 km/s, $\chi^2_\nu = 0.965$ — pass.
- Mannheim: RMS 4.694 km/s, $\chi^2_\nu = 3.202$ — fail.

Mannheim therefore has the smallest unweighted RMS, while NFW has the better uncertainty-weighted score and AICc. This is one bounded galaxy protocol. It omits major systematics and provides neither population-level model selection nor a universal refutation.

4. Bateman–Turok and the failure of an all-field shortcut

The Bateman–Turok positive Euclidean lattice and hidden ghost-parity proposal raises a natural deterministic question: can the complete action gradient be bounded below at the free bilaplacian scale for every positive field?

Paper 22 answers **no**. On $L_n = n^{24}$, an exactly mean-compensated critical source is inverted with the periodic graph Green operator, lifted above its minimum, and given a small four-way coupling. The resulting fields are positive, nonseparable, and polynomial-contrast. Their action stays bounded above and away from zero, while

$$\frac{Q_n}{\omega_{L_n}^2} = O(n^{-2}) \rightarrow 0.$$

This refutes one deterministic all-field scaled-PL architecture. It does not refute the full Bateman–Turok programme. A successful route could still use Gibbs typicality, the actual interacting Witten or H^{-1} structure, or a different reconstruction argument.

5. Reverse foundations and theory assemblies

Papers 20–21 and the interactive atlas ask whether changing the mathematical assumptions changes what physics can be built. The axes include classical, choice-weakened, constructive, internal/topos, and finite regimes; finite, Hilbert, Krein, algebraic, smooth/distributional, and localic carriers; and sixteen physical obligations from states and dynamics through renormalization, reconstruction, and empirical adequacy.

The atlas contains nine programme prototypes. The mainstream GR/QFT reference has direct records for 16/16 jobs, while the Mannheim and pure-Weyl prototypes each show 15/16 and other programmes less. Those numbers are coverage envelopes, not rankings. A prototype may select different models and carriers for different jobs; no result follows until their interfaces are proved.

Eight theory passports instead follow six stage functions—foundations, state space, dynamics, observable, prediction, and empirical benchmark. Four reach bounded data: GR/Cassini and GR+NFW/NGC 3198 pass their declared gates; Newtonian baryons and Mannheim/NGC 3198 fail theirs. Bateman–Turok, the free Krein mode, the constructive coded wave, and the pure-Weyl BV route have not yet reached an empirical test.

How to review the work

Start with the [Reverse Physics Atlas](#):

1. **Dimensions guide** explains every axis in non-specialist language.
2. **Theory profiles** shows obligation coverage without pretending it is empirical validity.
3. **Assemblies** separates bounded model tests, research programmes, and interface/calibration ledgers.
4. **Theory journeys** exposes each end-to-end passport and its first blocker.
5. **Weyl BV routes** shows the eleven-gate causal and quantum frontier.
6. **Evidence** exposes certificates, literature records, and the two Lean/Physlib proof passports.

For a technical claim, follow

```
claim
  → producer
  → machine-readable certificate
  → independent verifier
  → adversarial or mutation test
  → paper with explicit non-claims
```

Re-running a producer is reproduction, not independent verification. Exact and numerical evidence remain different types. A timeout or skip is never a pass.

Paper map

- **Papers 01–06:** fourth-order completion, field representations, and interaction obstructions.
- **Papers 07–08:** pure-Weyl residual cohomology and covariant causal transport.
- **Papers 09–13:** clocks, compact phase spaces, gravity–light transfer, anomaly, and nonlinear tangent cones.
- **Papers 14–18:** black-hole architecture, Phase-1 synthesis, endpoint non-selection, defective resonance, and static thermodynamics.
- **Paper 19:** assumptions behind the Mannheim rotation-curve explanation and the bounded NGC 3198 check.
- **Paper 20:** whether gauge algebra is input, derived structure, or representation-dependent.
- **Paper 21:** the reverse-foundations programme, literature synthesis, atlas, and evidence appendices.
- **Paper 22:** the compensated Green-tail counterfamily and deterministic BT torus no-go.

Highest-leverage next questions

1. Is the certified BRST–Hadamard pseudo-state positive on the declared physical cohomology, or can positivity be exactly obstructed?
2. Can Lorentzian renormalized products and a QME be constructed without crossing the dependency boundaries?
3. Is the BT Green-tail family probabilistically negligible, or does it obstruct the actual interacting H^{-1} or Witten estimate?
4. Which assembly interfaces can be certified so that coverage becomes an end-to-end prediction rather than a portfolio?
5. Do the NGC 3198 model orderings persist under systematics, other galaxies, held-out tests, and physically constrained priors?
6. Can the black-hole double pole be promoted to a complete causal waveform with a real source and detector observable?
7. Which mathematical assumptions are genuinely necessary, rather than merely inherited from a convenient representation?

Bottom line

The programme has not shown that Weyl gravity or Bateman–Turok physics is a viable description of nature. It has also not reduced their questions to one generic “ghost” objection.

Its strongest durable result is methodological and technical:

Physical postulates, mathematical carriers, logical strength, causal and quantum lifecycle states, interface proofs, and empirical gates can be separated—and each separation exposes a concrete theorem or missing bridge that experts can inspect.

That makes the repository a research instrument, not a verdict machine.